

Question B1

This question carries 17 marks in total.

(a) [5 MARKS] Consider the conical surface, S , defined by

$$z = 4\sqrt{x^2 + y^2} \quad \text{for} \quad -1 \leq x, y \leq 1.$$

Show that the normal to S , pointing *out* of the cone, can be written as:

$$\mathbf{n} = \frac{4x}{\sqrt{x^2 + y^2}} \mathbf{i} + \frac{4y}{\sqrt{x^2 + y^2}} \mathbf{j} - \mathbf{k}.$$

(b) [3 MARKS] Given the same surface as in part (a), show that S can be parameterized as follows:

$$\mathbf{r}(s, t) = s \cos(t) \mathbf{i} + s \sin(t) \mathbf{j} + 4s \mathbf{k}, \quad \text{for} \quad 0 \leq s \leq 1, 0 \leq t \leq 2\pi.$$

(c) [3 MARKS] Let

$$\mathbf{F} = y\mathbf{i} - x\mathbf{j} + 5\mathbf{k}.$$

Calculate the flux of $\mathbf{F}$ *out* of the cone. i.e. evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where S is the same surface as in part (a) and $d\mathbf{S}$ is the vector element of area pointing out of the cone.

(d) [6 MARKS] Let $\mathbf{r}(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + 5t \mathbf{k}$ define a helix C for $0 \leq t \leq 2\pi$.

Calculate

$$\int_C \mathbf{F} \cdot d\mathbf{r}, \quad \text{for} \quad \mathbf{F} = z\mathbf{i} - x\mathbf{j} + y\mathbf{k}.$$

You may find the identity $\cos^2(A) = \frac{1}{2}(1 + \cos(2A))$ useful.

Solutions

Answer (a):

Let the surface be defined implicitly as $f(x, y, z) = z - 4\sqrt{x^2 + y^2} = 0$.

The normal vector is given by the gradient of f :

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

Calculating the partial derivatives:

$$\frac{\partial f}{\partial x} = -4 \cdot \frac{1}{2\sqrt{x^2 + y^2}} \cdot 2x = -\frac{4x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial f}{\partial y} = -4 \cdot \frac{1}{2\sqrt{x^2 + y^2}} \cdot 2y = -\frac{4y}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial f}{\partial z} = 1$$

$$\text{So, } \nabla f = -\frac{4x}{\sqrt{x^2 + y^2}}\mathbf{i} - \frac{4y}{\sqrt{x^2 + y^2}}\mathbf{j} + \mathbf{k}.$$

This normal vector points "inward" (in the positive z -direction, into the cone). To obtain the normal pointing *out* of the cone, we negate it:

$$\mathbf{n}_{out} = -\nabla f = \frac{4x}{\sqrt{x^2 + y^2}}\mathbf{i} + \frac{4y}{\sqrt{x^2 + y^2}}\mathbf{j} - \mathbf{k}$$

Answer (b):

Using the provided parameterization $\mathbf{r}(s, t) = s \cos(t)\mathbf{i} + s \sin(t)\mathbf{j} + 4s\mathbf{k}$:

Let $x = s \cos(t)$ and $y = s \sin(t)$.

Then, $x^2 + y^2 = s^2 \cos^2(t) + s^2 \sin^2(t) = s^2(\cos^2(t) + \sin^2(t)) = s^2$.

Since $0 \leq s \leq 1$, we have $\sqrt{x^2 + y^2} = s$.

From the parameterization, the z -component is $z = 4s$. Substituting s , we get $z = 4\sqrt{x^2 + y^2}$.

This matches the equation of the cone from part (a). The parameters represent polar coordinates where s is the radius and t is the angle. The ranges $0 \leq s \leq 1$ and $0 \leq t \leq 2\pi$ sweep out the surface of the cone over the unit disk $x^2 + y^2 \leq 1$.

Answer (c):

To calculate the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ out of the cone, we use the parameterization from (b):

$$\mathbf{r}_s = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + 4\mathbf{k}$$

$$\mathbf{r}_t = -s \sin(t)\mathbf{i} + s \cos(t)\mathbf{j}$$

The vector element of area is $\mathbf{r}_s \times \mathbf{r}_t$:

$$\mathbf{r}_s \times \mathbf{r}_t = -4s \cos(t)\mathbf{i} - 4s \sin(t)\mathbf{j} + s(\cos^2(t) + \sin^2(t))\mathbf{k} = -4s \cos(t)\mathbf{i} - 4s \sin(t)\mathbf{j} + s\mathbf{k}$$

This vector has a positive z -component ($s > 0$), which means it points inward. The outward pointing vector element of area is $d\mathbf{S}_{out} = -(\mathbf{r}_s \times \mathbf{r}_t)dsdt$:

$$d\mathbf{S}_{out} = (4s \cos(t)\mathbf{i} + 4s \sin(t)\mathbf{j} - s\mathbf{k})dsdt$$

On the surface, $\mathbf{F} = s \sin(t)\mathbf{i} - s \cos(t)\mathbf{j} + 5\mathbf{k}$.

$$\mathbf{F} \cdot d\mathbf{S}_{out} = [(s \sin t)(4s \cos t) + (-s \cos t)(4s \sin t) + (5)(-s)]dsdt = -5s dsdt$$

Integrating over the domain $0 \leq s \leq 1$ and $0 \leq t \leq 2\pi$:

$$\text{Flux} = \int_0^{2\pi} \int_0^1 -5s \, ds \, dt = \int_0^{2\pi} \left[-\frac{5}{2}s^2 \right]_0^1 dt = \int_0^{2\pi} -\frac{5}{2} dt = -5\pi$$

Answer (d):

Given $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + 5t\mathbf{k}$, the differential is:

$$d\mathbf{r} = (-\sin(t)\mathbf{i} + \cos(t)\mathbf{j} + 5\mathbf{k})dt$$

The vector field is $\mathbf{F} = z\mathbf{i} - x\mathbf{j} + y\mathbf{k}$. Substituting $x = \cos(t)$, $y = \sin(t)$, $z = 5t$:

$$\mathbf{F} = 5t\mathbf{i} - \cos(t)\mathbf{j} + \sin(t)\mathbf{k}$$

The dot product is:

$$\mathbf{F} \cdot d\mathbf{r} = (5t)(-\sin(t)) + (-\cos(t))(\cos(t)) + (\sin(t))(5) = -5t \sin(t) - \cos^2(t) + 5 \sin(t)$$

Now integrate from 0 to 2π :

1. $\int_0^{2\pi} -5t \sin(t) \, dt$: Using integration by parts ($u = t, dv = \sin(t)dt \implies du = dt, v = -\cos(t)$):

$$= -5 \left([-t \cos(t)]_0^{2\pi} - \int_0^{2\pi} -\cos(t)dt \right) = -5 \left(-2\pi \cos(2\pi) + 0 + [\sin(t)]_0^{2\pi} \right) = 10\pi$$

2. $\int_0^{2\pi} -\cos^2(t) \, dt$: $\int_0^{2\pi} -\frac{1}{2}(1 + \cos(2t)) \, dt = -\frac{1}{2} \left[t + \frac{1}{2}\sin(2t) \right]_0^{2\pi} = -\pi$

3. $\int_0^{2\pi} 5\sin(t) \, dt$: $\left[-5\cos(t) \right]_0^{2\pi} = -5(1 - 1) = 0$

Summing these up: $\int_C \mathbf{F} \cdot d\mathbf{r} = 10\pi - \pi + 0 = 9\pi$.